International Journal of Scientific Research in Computer Science, Engineering and Information Technology
ISSN : 2456-3307
OPEN ACCESS
Available Online at : www.ijsrcseit.com
doi : https://doi.org/10.32628/CSEIT2410410

Smart Car Parking System

Sumathi A[1], Lakshmi Bhaskar[2], Prateeksha K Ankolekar[3], Shreya Shenoy[3], Vaishnavi Bhatt[3]
[1]Associate Professor, [2]Assistant Professor, [3,4,5]Student
Department of Electronics and Communication Engineering, BNM Institute of Technology, Bengaluru, India

ARTICLEINFO

Article History:

Accepted : 10 Aug 2024
Published: 30 Aug 2024

Publication Issue
Volume 10, Issue 4
July-August-2024

Page Number
312-316

ABSTRACT

In this fast-growing economy, the number of vehicle users is growing tremendously, wasting a significant amount of time and fuel while searching for an available parking space. This work highlights the optimum utilization of parking slots without the physical intervention of humans by making use of an IoT-integrated embedded system. This study aimed to create a mobile application for car parking that uses NodeMCU as the Microcontroller, which is further integrated with Radio-Frequency Identification (RFID) and the Internet of Things (IoT) to detect available parking slots, saving people's time. When combined with an interactive mobile application, this technology provides an efficient solution by highlighting the availability and occupancy of parking spaces as green and red, respectively. It contributes to enhancing the quality of life of people living in congested metropolitan cities by digitizing the conventional parking system.

Index Terms - Internet of Things, Radio-Frequency Identification, FASTag, Parking premises, Car Parking System

I. INTRODUCTION

In today's fast-paced world, the exponential growth in population has resulted in a steady increase in the number of four-wheelers. This has resulted in inefficient management of accessible car parking slots. With a population of over 1.5 billion people and the third-largest road network in the world, India is home to close to 295.8 million automobiles overall in the fiscal year 2019. Furthermore, more than 60% of the population in India appears to use personal or shared vehicles for commuting, suggesting that road travel is the primary mode of transportation in this country. In light of these figures, we estimate that the automobile saturation rate is roughly 21%. The complexity of the project is intermediate from the developer's perspective to make it easy to use from the users' perspective.

The task of finding a parking space to park one's vehicle is a prevalent problem all over the world. This issue is aggravated in those places where a large number of people visit to get services of government offices or public places like shopping malls, hospitals etc., end up having no proper system for parking cars. A system is needed to identify the availability of parking space and inform the person who is in need of it at that instant of time. This increases the efficiency of the parking system where a large number of people move in and out of the parking space which has no limits on time, reduces traffic congestion near public places and also the time of the people in searching for available parking slots. The proposed system is designed in such a way that it effectively reduces traffic congestion near shopping malls and mismanagement of parking space by allowing the user to know the available parking slot before a person enters a mall across cities.

A live visual representation of the parking lot is provided via the smartphone application. In line with the current time scenario, it continuously updates and displays the status of the slot's availability and occupancy. Manual payment systems that require a lengthy wait near the departure gate were originally the norm for parking. This system automates that by taking money directly out of the wallet and allocating it in accordance with the time duration of the slot occupancy. This project can be implemented on a big scale, spanning an area of more square meters, with its applicability not being restricted to a single location or locality. Additionally, efficiency and dependability remain unaffected by the length of time or the number of hours that a parking space is used.

II. LITERATURE SURVEY

In today's world, parking issues have grown common in every metropolitan area and are escalating at an alarming rate. As a result, parking contributes to traffic congestion. Parking cars poses several challenges, including managing the number of vehicles parked and monitoring vehicle entry and exit from the parking zone. Using the concept of the Internet of Things, this paper proposes a solution to these problems. The proposed system accurately monitors and updates the check-in and check-out details of the vehicle approaching the car park. It also highlights the total count of cars and the available vacant slots within the car park on the screen. [4]

An alternative method describes a way for tracking and regulating a car park using an efficient parking management system driven by an Arduino UNO. The main goal of this system is to optimize the time taken to spot a vacant parking slot in turn reducing the consumption of fuel. Analogous to the previous system, number of cars in the car park are noted and vacancies are monitored. A sensor is used to locate unoccupied parking slots. Problems caused by the lack of a dependable, efficient and modern parking system are eradicated, aiding to the enhancement of the project feasibility.

An additional system has been developed that provides the users' with the feature to book vacant parking slots from any location with the help of a mobile application. Users' are also navigated to the parking slot they booked using the mobile application and also can make the payment online. Sensors present at the exit gate will monitor the vehicle's payment details. In case of discrepancies the user can pay the due amount through online modes of payment. [5]

Furthermore, a system has been devised using a variety of sensor circuitry and the cloud (server). It is an effective parking method that outperforms traffic congestion. This project has been subsequently developed into a smart parking system with automatic billing and a fully automated system employing layered parking. Safety procedures include driver face recognition and car number tracing. By giving each person a different OTP and ensuring that they park in the designated slot, further precautions have been taken to prevent malfunctions caused by the wrong vehicle entering the slot. [8]

III.RESEARCH METHODOLOGY

3.1 Theoretical framework

NodeMCU, an open-source platform with built-in Wi-Fi connectivity, drives this circuit, facilitates the connection between devices, and aids in data transmission and communication. Every user availing the services provided by this proposed system will be required to register with the agency, will have to take an RFID Tag analogous to the FASTag currently in use. Each RFID Tag will have a unique ID that the users should use to log into their account on the Smart Car Parking system application on their mobile. This Android application will display the availability of parking slots. Green is used to highlight vacant slots, in contrast to Red which displays occupied slots.

The RFID Tag affixed to the vehicle's windshield is scanned by the RFID EM -18 reader module installed near the parking premises as soon as the car approaches the proximity of the entry gate. The parking setup includes an IR sensor that detects the presence of the car at the entry gate, furthermore, it prompts the RFID EM-18 reader, installed near the entry gate, to scan the RFID Tag affixed to the vehicle's windshield subsequently the car approaches the parking premises. Following this, the servo motor drives the gate to open, permitting the vehicle within the parking area. the time of entry is noted in the real-time database. The vehicle can be parked in the vacant slot as displayed in the mobile application.

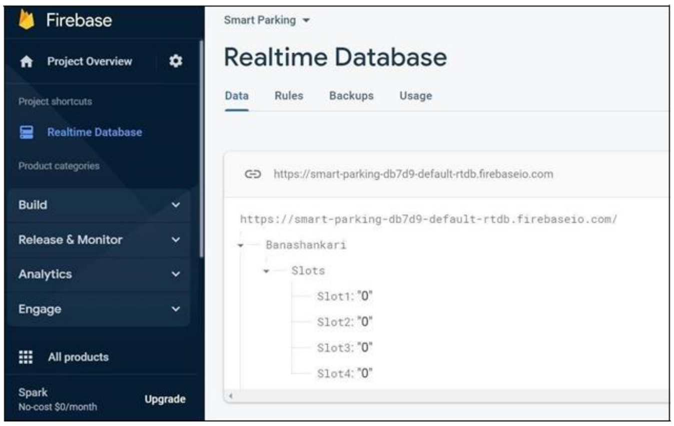
The parking slot setup includes an IR sensor which is used to detect the occupancy of that particular slot. This data is sent to the real-time database, which updates the mobile application accordingly. While checking out from the parking area the same procedure is followed, the time of exit is noted and the parking fare is calculated based on the hours of usage. This amount is then prompted on the app and can be debited from the user's account using one of the various payment methods available on the mobile application.

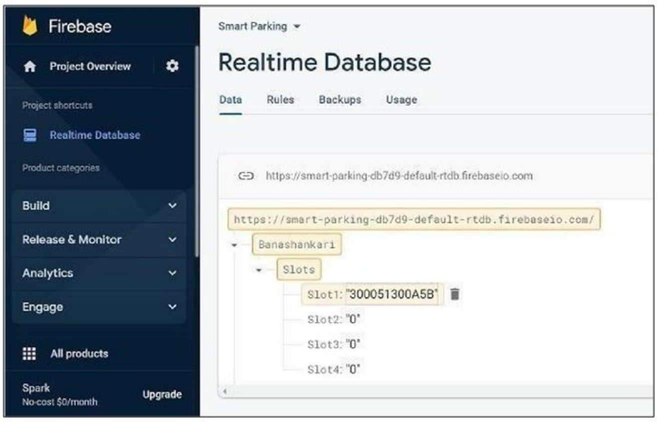
3.2 Implementation

When a car is detected by the IR sensor and the RFID card being read is valid, a servo motor opens the entrance gate, allowing the car to occupy the vacant slot. After checking the parking slot's availability on the mobile app, the vehicle occupies the vacant slot. A smartphone application to check the availability of parking spaces in various locations is being developed using the Integrated Development Environment (IDE) tool Android Studio. Only the RFID tag will allow the user to access this application.

When the user logs into the mobile application, the authentication occurs when the user enters their RFID card number and the password associated with the same account. The app redirects the user to a page where one can check the slot availability while parking. Firebase by Google, a cloud-hosted database, was utilized as the project's backend to record and update the real-time occupancy and availability of slots under each person's unique RFID card value. The slot in the mobile app is represented with green color indicating vacancy, and stored as “0” in the real-time database. It turns red when the slot is occupied and subsequently updates the real-time database as “1”. When the respective car vacates the parking slot, it is simultaneously updated as "0" in the real-time database to indicate that the parking spot is free.

Furthermore, the user can make the payment while exiting the parking premises depending on the hours of usage. If the user exits the parking premises before the expiry of a designated time period set by the organization employing this facility; they’ll be charged a fixed base price for the same. Additional charges are assessed on an hourly basis when the user exceeds the allotted time.





IV. RESULTS AND DISCUSSION

The implementation of the Smart Car Parking System aims to address the issue of insufficient parking slots and improve the overall parking experience. By continuously monitoring the system, information about empty and occupied slots is updated in real-time, allowing users to find available parking spaces efficiently. The system utilizes RFID cards and IR sensors to automate the process of parking slot allocation. This reduces the time and effort spent by users in searching for parking spots. Additionally, it helps in minimizing fuel wastage, contributing to a greener environment. The integration of the mobile application provides users with a convenient way to check the availability of parking spaces. The application uses Firebase as the backend to ensure real-time updates and accurate slot information. Furthermore, the system incorporates an integrated payment feature, allowing users to make payments while exiting the parking premises based on the hours of usage. This digitizes the payment process and provides a seamless experience for the users.

Future Scope:

The Smart Car Parking System has great potential for further enhancements. Some possible improvements and additions include:

Spot Reservation: Introduce the option for users to reserve a parking spot in advance through the mobile application. This feature would ensure a guaranteed parking space upon arrival.

GPS Tracking: Incorporate a GPS tracking system in the mobile application to navigate users to available vacant parking slots. This feature would provide real-time directions and guidance to the nearest available parking spot.

Enhanced Security: Implement additional security measures such as CCTV surveillance and license plate recognition to further enhance the safety and protection of the parking premises.By incorporating these enhancements, the Smart Car Parking System can further improve the overall user experience and address additional challenges related to parking availability and convenience.

In conclusion, the implementation of the Smart Car Parking System demonstrates the effectiveness of IoT-based Android mobile application technology for car parking. The system offers numerous benefits, including time and fuel savings, increased protection, and a digitized payment process. With future enhancements, this project has a promising scope to revolutionize the parking experience and improve people's quality of life.

V. REFERENCES

[1]. Shashank Shenoy Basty , , Rajeshwari Kiwad , Srikanth Vittal , "Mohammad Moin Ullah, 2022, RFID based Smart Parking System", INTERNATIONAL JOURNAL OF ENGINEERING RESEARCH & TECHNOLOGY (IJERT) Volume 11, Issue 07 (July 2022).

[2]. Karimi, K., & Atkinson, G. (2013). "What the Internet of Things (IoT) needs to become a reality". White Paper, FreeScale and ARM, 1-16.

[3]. Rahman Atiqur. iPark: Automated smart parking system. International Journal of Advances in Applied Sciences (IJAAS), p-ISSN 2252-8814, e-ISSN 2722-2594.

[4]. G. Naresh, Dr. B. Indira (2017). Smart City – Parking Management Using Mobile Application. International Journal of Advance Research in Science and Engineering. Bol. nP. 6, Special Issue(01), September 2017, BVCNCS 2017.

[5]. Idris, M. Y. I., Leng, Y. Y., Tamil, E. M., Noor, N. M., & Razak, Z. (2009). "Car park system: a review of smart parking system and its technology." Information Technology Journal, 8(2), 101-113.

[6]. Ahteshamul Huq Osmani, Ashwini Gawade, Minal Nikam, Swati Wavare, Research paper on "Smart City Parking System", Vol-2 Issue 3 2016

[7]. Shruthi Mudaliar, Shreya Agali, Sujay Mudhol, Chaitanya K Jambotkar, "IoT Based Smart Car Parking System" Volume 5 Issue 1, JANUARY 2019, ISSN: 2395-1052.

[8]. Z. Pala and N. Inanc, "Smart Parking Applications Using RFID Technology," 2007 1st Annual RFID Eurasia, Istanbul, Turkey, 2007, pp. 1-3, doi: 10.1109/RFIDEURASIA.2007.4368108.